Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Miscellaneous topics

Image and Signal Processing

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Supplementary topics

…for frequency analysis

Frequency bands

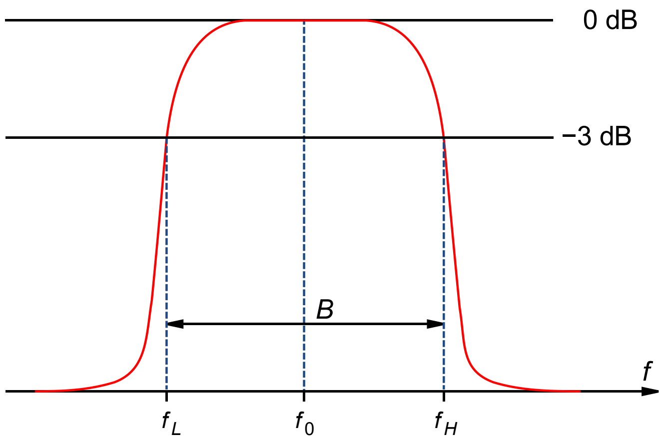
- A signal is called **band-limited** if its frequency components are limited within frequencies:

$$f_L \leq f \leq f_H$$

- The **bandwidth** of a signal is defined as the difference between the extreme frequencies:

$$B = \Delta f = f_H - f_L$$

- A strictly band-limited signal does not carry energy at frequencies outside the band limits
- In practice, a signal is considered band-limited if its energy outside of a frequency range is low enough to be considered negligible



Amplitude spectrum of a bandlimited signal.
Source of illustration: Wikimedia

The decibel scale

Alternative scale 1: The decibel scale is a logarithmic scale applied to the power spectrum (typically the vertical axis), where the power spectrum describes how the signal's energy is distributed across frequencies.

- Decibels (dB) are often used to measure the power of a signal relative to some reference level.
- Formally, a **bel** is defined as a logarithmic ratio between two power quantities, a measured and reference value:
- The power carried by a sinusoidal signal is proportional to the square of the signal amplitude, so one can also write: (c is a proportionality constant)
- Decibels are commonly used in **power spectra**, which are like amplitude spectra, but use a decibel scale.
- Example:
  - Acoustic signals = pressure waves traveling through a medium
  - The standard reference amplitude used for acoustic signals $A_{ref} = p_0$ is 20 µPa, which corresponds approximately to the quietest sound that a person can hear.

$$1\mathrm{B} = \log_{10}\left(\frac{P_{meas}}{P_{ref}}\right)$$

$$1\mathrm{dB} = 10\mathrm{B} = 10\log_{10}\left(\frac{P_{meas}}{P_{ref}}\right)$$

$$= 10\log_{10}\left(\frac{cA_{meas}^2}{cA_{ref}^2}\right)$$

$$= 20\log_{10}\left(\frac{A_{meas}}{A_{ref}}\right)$$

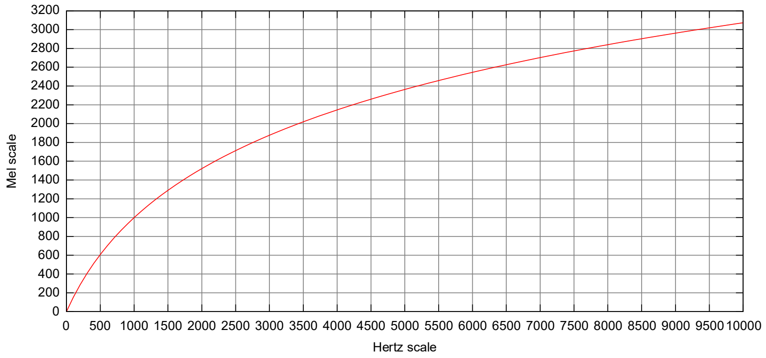
The Mel scale

Alternative scale 2: The Mel scale is a logarithmic-like scale applied to the frequency axis (usually horizontal), designed to reflect how humans perceive pitch.

- Main idea:
  - The Mel scale is a frequency scale designed to match **human auditory perception**.
  - Humans perceive frequency non-linearly, with higher sensitivity at low frequencies

- The Mel scale:
  - Non-linear spacing of frequency bins: higher resolution at low frequencies, lower resolution at high frequencies.
  - Maps frequencies from Hertz (Hz) to the Mel scale
  - **Usage**: Replace the linear frequency axis with the Mel scale to better reflect perceived sound

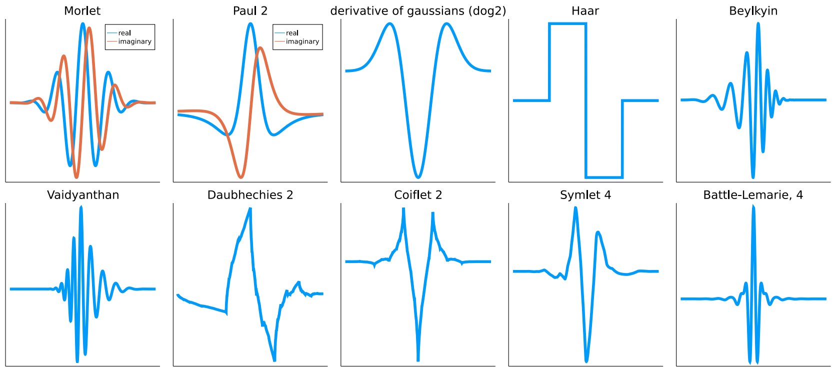
$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$



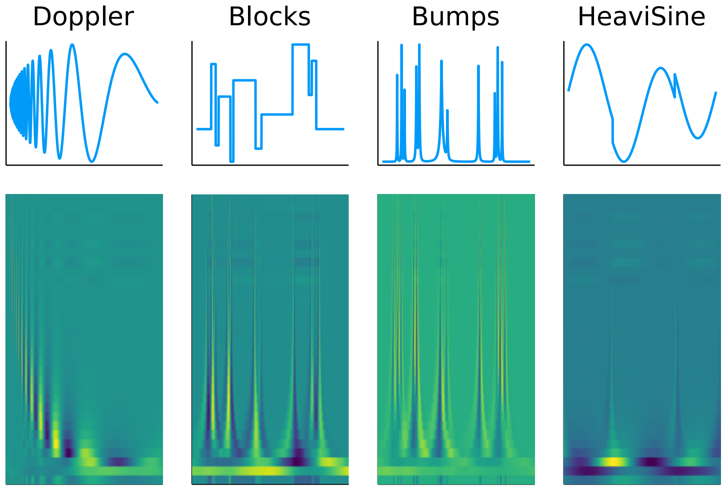
- Why use Mel-scaled spectrograms?
  - Closer to human hearing → Useful in speech/audio processing.
  - More compact representation → Reduces unnecessary high-frequency details.

Wavelet decomposition

- Problems with Fourier transform:
  - Fourier representation: Functions are expressed as linear combinations of sinusoidal functions.
  - Sinusoids: They have infinite extent in the time domain – they have no distinct start or end.
  - Localization issue: The Fourier transform provides frequency information but lacks time localization
  - Limitations: It is not well-suited for capturing transient features or localized events in a signal.
- Solution: Wavelet transform
  - Introduce new basis functions that are localized in both time and frequency domains.
  - Decompose a signal using scaled and shifted versions of these basis functions.
  - Advantage: Wavelets provide good frequency resolution while maintaining good time localization.



Exemplary mother functions for Wavelets, from which the base functions are derived by scaling in x- and y-direction.



Wavelet decomposition result: Like an STFT representation, but achieved without requiring multiple transformations.

Compression

Compression

- Signal compression refers to the process of reducing the size (in bytes) of a signal while preserving essential information.
- Goal:
  - Eliminate redundancy or irrelevant information.
  - Trade-off: Compression ratio / information loss / method complexity
- Methods:
  - **Lossless compression**: Exact reconstruction is possible
    - Run-length encoding (RLE)
    - Huffmann coding
    - Lempel-Ziv-Welch (LZW)

Example: Run length encoding (RLE).

Input:

```
WWWWWWWWWWWWBWWWWWWWWWW
WWWBBBWWWWWWWWWWWWWWWWW
WWWWWWWWBWWWWWWWWWWWWWW
```

Output: Compressed data

```
12W1B12W3B24W1B13W
```

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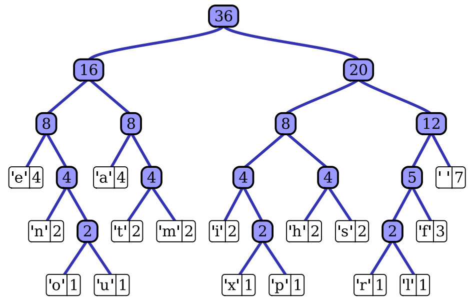
Example: Huffmann coding.

Idea: Use a variable-length code for different tokens (or byte sequences), assigning shorter codes to the most frequent tokens.

Input: “this is an example of a huffman tree”. Tokens by frequency:



Output: Huffmann code table

Char	Freq	Code	Char	Freq	Code
space	7	111	s	2	1011
a	4	010	t	2	0110
e	4	000	l	1	11001
f	3	1101	o	1	00110
h	2	1010	p	1	10011
i	2	1000	r	1	11000
m	2	0111	u	1	00111
n	2	0010	x	1	10010

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  - **Lossy Compression**: Higher compression ratios by discarding some signal information. Exact reconstruction is not possible.
    - Subsampling / down-sampling
    - Transform coding (e.g., discrete cosine transform, discrete wavelet tr.)
  - **Hybrid compression**: Lossless + lossy. Examples: MP3, JPEG...

Example: Lossy data compression with AI-based signal reconstruction. Reference: Link

Data: PPG sensor data (top), after SZ compression (middle), and after enhanced reconstruction (bottom)